



RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

Bachelor of Science in Applied Sciences
Second Year - Semester II Examination - November/December 2023

MAP 2202 - REAL ANALYSIS II

Time: Two (02) hours.

Answer ALL (04) questions.

1. a) Let $a_n = \frac{3n^2+5n+4}{2n^2+3}$ for all $n = 1, 2, 3, \dots$.

Find the limit $\lim_{n \rightarrow \infty} a_n$ and hence determine whether or not the infinite series $\sum_{n=1}^{\infty} a_n$ is convergent.

(25 marks)

- b) Using the comparison test, show that the series $\sum_{n=1}^{\infty} \frac{1}{n^n}$ is convergent.

(25 marks)

- c) Using the integral test, show that the series $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^\alpha}$ is convergent if $\alpha > 1$ and is divergent if $0 < \alpha \leq 1$.

(25 marks)

- d) Determine whether the following series is convergent or divergent:

$$\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{2021n + 2023}{2024n - 2022} \right)^n$$

(25 marks)

- a) Determine the radius of convergence and interval of convergence of the power series:

$$\sum_{n=1}^{\infty} \frac{(n+1)}{(n+2)(n+3)} x^n.$$

(50 marks)

- b) Prove the followings:

$$\begin{aligned} \text{i.} \quad & \frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1} \quad \text{if } |x| < 1, \\ \text{ii.} \quad & \ln(1-x) = -\sum_{n=1}^{\infty} \frac{x^n}{n} \quad \text{if } -1 \leq x < 1. \end{aligned}$$

(50 marks)

- a) Evaluate the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{y \sin(x^2 + y^2)}{x^2 + y^2}.$

(20 marks)

- b) Suppos $z = e^y \cos x$, $x = s^2 t$, and $y = st^3$. Find $\frac{\partial z}{\partial t}$.

(20 marks)

- c) Determine all critical points of the function $f(x, y) = x^3 + y^3 - 3xy$.

Discuss the nature of each critical point.

(60 marks)

Use the method of Lagrange multiplier to solve the following problems:

- a) Find the minimum value of the function $f = x^2 + y^2 + z^2$ subject to $2x + 3y - z = 1$.

(60 marks)

- b) Find the largest product of the numbers a and b when $a^2 + b^2 = 1$.

(40 marks)

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